

Hybrid Edge AI Node: A Fractal Governance Architecture for Sovereignty-Aware Distributed AI Processing

Jakkrit Boonma

Independent Researcher · jaggid.boonma@yahoo.co.th · +66 081-007-4833

Abstract

Centralized cloud architectures increasingly struggle to meet the latency, data-sovereignty, and bandwidth demands of industrial AI workloads. This paper presents Hybrid Edge AI Node, a distributed processing architecture that unifies six previously separate research threads — hierarchical edge orchestration, hierarchical federated aggregation, human-in-the-loop AI governance, distributed leader election, swarm-based knowledge sharing, and sovereign edge computing — into a single, organizationally scalable architecture. The design rests on four core layers and six pillars, over which a fractal governance hierarchy (Primary, Secondary, and Zone Master Nodes) can grow or contract to match an organization's size without redesign. A three-state Policy Boundary Escalation mechanism governs every autonomous decision — auto-approved within policy, halted pending human approval outside it, or executed immediately with retrospective notification when system survival is at stake. A Role Transformation protocol, guarded by a generation-number fencing scheme, allows Worker Nodes to assume Master responsibilities during failures without producing split-brain conditions. A Sovereignty-Aware Swarm Broadcast algorithm allows the network to learn collectively while guaranteeing that only distilled, customer-free insights cross jurisdictional borders. We describe seven core algorithms in full, present a real-world case study (CineNexus Pro, a cloud-based AI media studio), and situate the architecture against seventeen external works. The contribution is not any single mechanism in isolation — each has precedent — but their integration into one coherent, governable, and organizationally scalable architecture, which to our knowledge has not previously been reported.

Keywords: Edge Computing, Distributed AI, Fractal Governance, Human-in-the-Loop, Leader Election, Swarm Intelligence, Data Sovereignty, Zero-Trust Security

1. Introduction

The rapid growth of industrial AI has produced workloads that traditional centralized cloud architectures struggle to serve. Two problems recur across manufacturing, media, healthcare, and public-sector deployments: response latency incurred by round-tripping data to a distant data center, and legal exposure created by moving sensitive data across organizational or national boundaries. Edge computing addresses the first problem by moving computation close to the data source; however, most existing edge frameworks assume a fixed two-tier topology (cloud and edge) and provide no mechanism for an organization's control structure to grow as the organization itself grows internationally.

This paper addresses a narrower and, we argue, more tractable question than "how should edge AI be architected in general": given an organization that starts as a single site with one control node and a handful of processing nodes, how should its control-plane architecture evolve as it expands to

multiple sites, multiple countries, and eventually a global footprint — without requiring a redesign at each stage, and without weakening human oversight of increasingly autonomous decisions? We answer this question with Hybrid Edge AI Node, an architecture built on four layers and six pillars, in which control-plane nodes are recursively self-similar (fractal): every level above a Worker Node is a Master Node, differing only in the geographic or organizational scope it governs.

The remainder of this paper is organized as follows. Section 2 situates the work against six areas of prior research. Section 3 presents the canonical architecture. Section 4 introduces the fractal governance model, including the Policy Boundary Escalation and Role Transformation mechanisms. Section 5 specifies the seven core algorithms. Section 6 describes the data lifecycle and sovereignty model. Section 7 presents a case study. Section 8 discusses limitations, and Section 9 concludes.

2. Related Work

Six areas of prior work bear directly on this architecture, each addressing one dimension of the problem in isolation.

2.1 Hierarchical Edge Orchestration

KubeEdge, a CNCF project, extends Kubernetes-native container orchestration to edge hosts through a three-tier split of cloud, edge, and device layers, with an edge runtime footprint of roughly 70 MB [1][2]. AWS IoT Greengrass similarly allows devices to process data locally and continue operating during connectivity loss, synchronizing once the connection is restored [3]. A systematic review of cloud-native edge orchestration notes that commercial offerings from major cloud vendors tend to share a common weakness: limited cross-platform interoperability and a high degree of vendor lock-in [4]. Both KubeEdge and Greengrass assume a fixed two-tier topology; neither provides a mechanism for the control plane itself to grow additional tiers as an organization expands geographically.

2.2 Hierarchical Federated Aggregation

Edge-assisted Federated Learning (eFL) introduces a middle tier of regional aggregation servers between edge devices and a global server: each regional server aggregates local models within its region, and the global server aggregates only the regional models [5]. Time-effective hierarchical aggregation algorithms address the straggler problem created by heterogeneous device speeds [6]. This body of work establishes the technical precedent for our Zone-Secondary-Primary hierarchy, but is scoped to model training; it does not extend to organizational governance, role failover, or policy enforcement.

2.3 Swarm Intelligence and Digital Immune Systems

Swarm Computing describes networks of autonomous, cooperating devices that give rise to collective artificial intelligence at the edge of the network, analogous to specialized bees contributing to a shared goal [7]. The EU-funded HYPER-AI project operationalizes this idea through self-CHOP principles (self-configuration, self-healing, self-optimizing, self-protecting) for swarms of heterogeneous compute nodes [8]. In industry, Gartner's Digital Immune System concept — combining observability, chaos engineering, and automated remediation — projects an 80% reduction in downtime for organizations that adopt it by 2025 [9]. None of this work conditions

knowledge propagation on prior authorization from a human-governed policy layer, nor does it address cross-border data protection during propagation.

2.4 Distributed Leader Election

Leader election algorithms — Bully, Paxos, and Raft — solve the problem of selecting a coordinator among peer nodes when the incumbent fails, with Raft favored for its comparative simplicity over Paxos while preserving strong consistency guarantees [10][11]. These algorithms assume peer nodes of comparable capability; none addresses promotion across a capability gradient, such as a resource-constrained Worker Node assuming Master responsibilities.

2.5 Human-in-the-Loop AI Governance

A 2026 framework for governable enterprise AI execution halts tool invocation physically — not merely by logging — whenever a risk assessment exceeds a configured threshold, persisting the pending state across restarts until a human confirms [12]. The Autonomy Threshold concept formalizes a three-layer control pattern: an evaluation layer against configured conditions, an execution-control layer that allows, blocks, or pauses actions, and an escalation layer that routes decisions to a human when thresholds are exceeded [13]. Gartner's Human-AI Delegation Framework similarly defines seven levels of authority-sharing between human and machine [14]. This literature is very recent (2025–2026), general-purpose, and — critically — offers no exception for situations where waiting for approval would itself cause system failure.

2.6 Sovereign Edge Computing

Sovereign edge computing, discussed principally in the context of NATO-aligned organizations, denotes infrastructure that guarantees all data collection, processing, and storage remain within a defined national or allied boundary, eliminating external cloud dependency and shielding data from extraterritorial legal access [15]. Such systems are also built to remain fully operational in Disconnected, Denied, Intermittent, and Limited (DDIL) network conditions [16]. Zero-Trust Architecture is separately proposed as a foundation for digital sovereignty through least-privilege access, continuous verification, and identity-based control [17]. This literature enforces an all-or-nothing rule — raw data may never cross a border — with no mechanism for the network to still learn collectively from events observed in one jurisdiction.

2.7 Positioning

Table 1 summarizes the gap this paper addresses: each area above solves one dimension of the problem well, but none combines organizational scalability, human governance with an emergency exception, cross-capability failover, and sovereignty-aware collective learning into a single coherent architecture.

Hybrid Edge AI Node component	Closest prior work	Gap this paper closes
Fractal governance hierarchy	KubeEdge [1]; hierarchical FL [5]	Relative, growable tiering as org. scales
Policy Boundary Escalation	AgentRunner [12]; Autonomy Threshold [13]	Emergency exception (act-then-notify)

Role Transformation	Raft/Paxos/Bully [10][11]	Cross-capability promotion, not peer voting
Sovereignty-Aware Swarm Broadcast	Digital Immune System [9]; Swarm Computing [7]	Policy-gated, sovereignty-filtered propagation
Data Sovereignty model	Sovereign Edge Computing [15][16]	Two-tier: raw data local, insight only crosses

Table 1. Positioning of Hybrid Edge AI Node relative to prior work

3. Canonical Architecture

The architecture is defined by four layers, three of which remain constant regardless of organizational size, and one of which scales with it.

Layer	Function	Scales with org. size?
Management & Orchestration	Hosts the Policy layer; job scheduling, API gateway, security & key management	Yes — see Section 4
Edge Execution	Zero-Trust, in-memory task execution on Worker Nodes	No
Swarm Intelligence	Cross-node knowledge exchange (Section 5.7)	No
Network & Security	mTLS transport, encryption in transit and at rest	No

Table 2. The four canonical layers

Six pillars are distributed across these layers: Hierarchical Policy Governance, Orchestration Intelligence, and Plug-and-Play Provisioning sit in the Management & Orchestration layer; Zero-Trust Security and Self-Healing sit in Edge Execution; Swarm Intelligence constitutes its own layer. A worker-node spectrum spans from Edge Micro-Compute Nodes (Jetson-class hardware, sufficient RAM and a full Linux stack to support genuine Zero-Trust execution) through single-GPU workstations to multi-GPU farms. Resource-constrained sensor/actuator hardware (e.g., ESP32-class microcontrollers) is explicitly excluded from the Worker Node category: such devices cannot perform in-memory decryption and are treated as upstream data sources outside the trust boundary. Node tiering is resolved dynamically by querying free RAM and GPU memory at dispatch time rather than by fixed classification, since available capacity changes continuously with load.

4. Fractal Governance Model

4.1 The Hierarchy Is Relative, Not Fixed

Every tier above a Worker Node is a Master Node; tiers differ only in scope. A Zone Master governs a branch or district; a Secondary Master governs a country; a Primary Master governs the organization as a whole. When an organization's Primary Master eventually requires a global parent (a Global Primary Master), the former Primary is demoted, in name only, to a Secondary — its technical role is unchanged. This recursive property means that an organization's governance structure grows in depth exactly as fast as its organizational complexity does, without a redesign at any stage.

Tier	Scope governed
Global Primary Master	Worldwide (future state, once an organization operates globally)

Primary / Secondary Master	Home country — status is relative and can be demoted as above
Secondary Master	Each additional country the organization expands into
Zone Master	Each branch or region within a country
Worker Node	Processes jobs

Table 3. The fractal governance hierarchy

4.2 Policy Boundary Escalation

An organization's board of directors defines the boundary of autonomous action, combining numeric thresholds for quantifiable decisions with an allowlist of pre-approved categories for qualitative ones. Every proposed action that might fall outside this boundary is checked by a single, universally invoked sub-routine before it executes.

Situation	System behavior	Timing
Within policy	Execute automatically	Immediate
Outside policy, ordinary	Halt, notify the authorized approver, and wait	Before action
Outside policy, system survival (Emergency Exception)	Execute immediately, then notify the next authority	After action

Table 4. Policy Boundary Escalation — three states

***Within policy = fully autonomous. Outside policy, ordinary = halt and wait for a human.
Outside policy, system-threatening = act first, notify immediately after.***

Every decision, regardless of path, is written to an append-only audit log. This is the single sub-routine invoked by Role Transformation (as an Emergency Exception), by Sovereignty-Aware Swarm Broadcast (as an ordinary Knowledge Update), and by the Bounded Staging Checkpoint (for long-overdue paid content awaiting disposal).

4.3 Role Transformation and Split-Brain Prevention

A Worker Node may be promoted to Master when the incumbent Master's heartbeat is lost, becomes overloaded, is suspected of compromise, or is network-partitioned — all four are treated identically as System Survival, and promotion proceeds under the Emergency Exception described above, without waiting for approval. The candidate with the highest weighted stability score (uptime, heartbeat consistency, and inverse error history) is selected. Promotion increments a network-wide generation number by one. When the original Master reconnects, it compares its own last-known generation number against the network's current value; if lower, it accepts a Worker role immediately and unconditionally — there is no automatic fallback and no negotiation. This mechanism, directly analogous to the term number in Raft, prevents split-brain conditions that would otherwise arise if a partitioned Master reappeared believing itself still in charge.

5. Core Algorithms

The system executes seven core algorithms end-to-end. Algorithms 1–4 form the basic job lifecycle; Algorithms 5–7 are governance mechanisms invoked by, and shared across, the other algorithms and by each other.

#	Algorithm	Layer
1	Secure Job Ingestion	Management & Orchestration
2	Dynamic Job Dispatching	Management & Orchestration
3	Secure Execution & Staging	Edge Execution
4	Bounded Staging Checkpoint	Edge Execution (background)
5	Policy Boundary Check	Cross-cutting
6	Role Transformation	Orchestration + Edge Execution
7	Sovereignty-Aware Swarm Broadcast	Swarm Intelligence

Table 5. The seven core algorithms (bold rows are given in full below)

5.1 Algorithms 1–4 (summary)

Secure Job Ingestion authenticates the client, issues a job-scoped AES-256 key from the key management system (KMS), classifies the job against the Policy layer to determine whether its output requires bounded staging (see Section 6), encrypts the payload immediately, and publishes it to a durable queue — the plaintext payload never persists outside this one step. Dynamic Job Dispatching polls the queue, filters candidate Worker Nodes by live heartbeat, queries each candidate's real-time free RAM and GPU memory (never a fixed tier), selects the least-loaded qualifying node, and dispatches under a short-lived lease to prevent duplicate assignment. Secure Execution & Staging decrypts the payload in memory only, executes the task, wipes the raw input unconditionally regardless of outcome, and then routes the output down one of three paths: awaiting assembly with sibling modules, encrypted bounded staging (Section 6), or immediate delivery-and-wipe. Bounded Staging Checkpoint runs as a scheduled background process that never deletes on a mere timeout; at 48 hours it applies an extended-storage fee and notifies the customer, and only escalates to Policy Boundary Check for a human disposal decision after a long-term threshold (e.g., 30 days) — permanently deleting paid content is never a fully automatic decision.

5.2 Algorithm 5: Policy Boundary Check

```

ALGORITHM: Policy_Boundary_Check
INPUT: proposed_action, policy_config

category = CLASSIFY(proposed_action, policy_config)

IF category == "SYSTEM_SURVIVAL":
    EXECUTE(proposed_action)
    NOTIFY(next_higher_authority, proposed_action, executed = TRUE)
    result = "EMERGENCY_EXECUTED"

ELSE IF category == "THRESHOLD_BASED" AND proposed_action.value WITHIN policy_config.threshold:
    result = "AUTO_APPROVED"
ELSE IF category == "ALLOWLIST_BASED" AND proposed_action.type IN policy_config.allowlist:
    result = "AUTO_APPROVED"
ELSE:
    PERSIST(proposed_action); NOTIFY(authorized_approver, proposed_action)
    result = WAIT_FOR_APPROVAL(proposed_action) # no automatic timeout

AUDIT_LOG.append({action: proposed_action, category, result, timestamp: NOW()})
RETURN result

```

5.3 Algorithm 6: Role Transformation

```

ALGORITHM: Role_Transformation
TRIGGER: heartbeat_lost OR overloaded OR breach_detected OR partition_detected

```

```

candidates = zone.get_online_workers()
FOR w IN candidates:
    w.stability_score = WEIGHTED_AVERAGE(uptime, heartbeat_consistency, INVERSE(error_count))
selected = MAX(candidates, key = stability_score)

Policy_Boundary_Check({type: "promote_to_master", node: selected}, category = "SYSTEM_SURVIVAL")

new_gen = ZONE_REGISTRY.get_generation() + 1
selected.promote_to_master(generation = new_gen)
selected.inherit_state(last_known_queue_state)
ZONE_REGISTRY.set_generation(new_gen)

ON original_master.reconnect():
    IF original_master.known_generation < ZONE_REGISTRY.get_generation():
        original_master.assign_role("worker") # no automatic failback

```

5.4 Algorithm 7: Sovereignty-Aware Swarm Broadcast

```

ALGORITHM: Sovereignty_Aware_Swarm_Broadcast
INPUT: discovery, origin_zone

IF NOT LOCAL_VALIDATE(discovery): RETURN
decision = Policy_Boundary_Check({type: "broadcast_new_rule", data: discovery}, category =
"KNOWLEDGE_UPDATE")
IF decision NOT IN ("AUTO_APPROVED", "EMERGENCY_EXECUTED"): RETURN

FOR w IN origin_zone.all_worker_nodes(): w.receive(discovery, form = "RAW")

distilled = STRIP_CUSTOMER_DATA(discovery)
distilled = STRIP_RAW_FILE_REFERENCES(distilled)
distilled = STRIP_NODE_IDENTIFIERS(distilled)
ASSERT distilled.contains_no_customer_data() == TRUE # mandatory checkpoint

primary_master.receive_upward(distilled) # via zone -> secondary -> primary
IF primary_master.validate_global(distilled):
    FOR zone IN primary_master.all_zones_worldwide():
        FOR w IN zone.all_worker_nodes(): w.receive(distilled, form = "DISTILLED")
        w.apply_rule(distilled)

```

6. Data Lifecycle and Sovereignty

Raw customer data is governed by Zero Data Retention without exception: decrypted only in memory, wiped unconditionally after use. Processed output that triggers data-protection or copyright obligations is instead governed by Encrypted Bounded Staging — held encrypted with a 48-hour checkpoint after which an extended-storage fee is applied and the customer notified, never silently deleted. This determination is made once, by the Policy layer, at ingestion, and is not revisited automatically.

Across borders, only two categories of information may leave the jurisdiction in which they were generated: aggregate business metrics, and distilled technical insight stripped of any customer or raw-file reference by the mandatory sovereignty filter in Algorithm 7. Raw data itself never crosses a border under any circumstance in the current design.

7. Case Study: CineNexus Pro

CineNexus Pro is a cloud-based AI media studio offering Media-as-a-Service (MaaS) through a proprietary payment unit (CN Token), currently operating a single-country baseline of one Master Node and two Worker Nodes (HP Z8 G4, Xeon Scalable, 128 GB RAM, RTX 3090/4090) connected by

a 10GbE core switch behind a Cloudflare edge (WAF, tunnel, R2 object storage). Total initial capital expenditure ranges from approximately 333,000 to 628,500 THB depending on GPU tier.

(THB)	Year 1	Year 2	Year 3
Total revenue (est.)	3,200,000	6,000,000	10,500,000
Net profit after tax	548,200	1,987,500	3,632,000

Table 6. CineNexus Pro three-year financial projection (business-plan estimate, not yet operationally validated)

As CineNexus Pro expands into additional countries, its Master Node — currently unambiguously Primary — is expected to remain Primary while newly established country deployments each become a Secondary Master, exercising the fractal hierarchy of Section 4.1 in practice for the first time. Cross-border reporting from these Secondary Masters back to the Primary is restricted to business metrics and swarm-distilled technical insight, in line with Section 6, keeping customer media content within the country in which it was uploaded.

8. Discussion

Two structural risks deserve explicit attention. First, Escalation Saturation: when the volume of requests routed to a human approver in the ordinary escalation path grows too large, operators tend to widen the automation threshold simply to reduce their own workload, an operational convenience decision rather than a considered governance decision, silently expanding the system's autonomous scope over time. The Policy layer's boundary configuration should therefore be reviewed on a fixed governance cadence, explicitly separated from any operational tuning performed for workload convenience. Second, capability mismatch during Role Transformation: because the Worker Node spectrum spans from Edge Micro-Compute Nodes to multi-GPU farms, a promoted Worker selected purely for stability may lack the hardware capacity to perform full Master-level orchestration; the current algorithm does not yet narrow the promoted node's responsibilities to match its available capacity.

More broadly, every algorithm in this paper is presented as pseudocode and has not yet been benchmarked against a production workload; the prototype implementation currently covers Algorithms 1–3 only in partial form, and Algorithms 4–7 remain unimplemented. Empirical validation — latency and throughput measurements under realistic load, and a live demonstration of Role Transformation and Sovereignty-Aware Swarm Broadcast — is the most consequential open item before this architecture can be assessed as more than a design proposal.

9. Conclusion and Future Work

Hybrid Edge AI Node integrates six previously disparate research threads into a single organizationally scalable architecture: a fractal governance hierarchy that grows with the organization, a three-state policy escalation mechanism with an explicit emergency exception, a generation-fenced role transformation protocol, and a sovereignty-aware swarm broadcast that separates local knowledge sharing from cross-border propagation. The contribution is the integration itself, evaluated in this paper against seventeen pieces of prior work spanning edge orchestration, federated learning, swarm

intelligence, distributed consensus, AI governance, and sovereign computing. Future work should prioritize, in order: (i) an empirical benchmark of the implemented algorithms under realistic load; (ii) full implementation of the governance algorithms (5–7), currently specified only as pseudocode; and (iii) a live, single-organization deployment — the CineNexus Pro case study described in Section 7 — as the basis for subsequent claims of broader industry applicability.

The empirical benchmarking methodology required for item (i) is one the author has applied before: an earlier peer-reviewed study by the same author compared block-access (iSCSI) and file-access (CIFS/Samba) I/O throughput across representative workloads — file transfer, web/mail, and database — using the Iozone, PostMark, and SQLio benchmark tools under controlled network conditions [18]. That study's finding that protocol performance is highly workload- and access-pattern-dependent, rather than uniformly favoring one design, directly motivates evaluating Hybrid Edge AI Node's algorithms under multiple realistic workload profiles rather than a single synthetic benchmark.

References

- [1] KubeEdge. "What is KubeEdge." KubeEdge Documentation. <https://kubedge.io/>
- [2] Cloud Native Computing Foundation. "KubeEdge, a Kubernetes Native Edge Computing Framework." Kubernetes Blog, 2019.
- [3] Amazon Web Services. "IoT Edge Computing." AWS Well-Architected Framework, IoT Lens.
- [4] "Cloud-Native Workload Orchestration at the Edge: A Deployment Review and Future Directions." PMC, National Center for Biotechnology Information.
- [5] "Edge assignment in edge federated learning." Discover Applied Sciences, Springer Nature, 2023.
- [6] "A Hierarchical Federated Learning Algorithm Based on Time Aggregation in Edge Computing Environment." Applied Sciences (MDPI), 13(9), 5821, 2023.
- [7] de Biase, L.C., Fedrecheski, G., Calcina-Ccori, P., Lopes, R., Zuffo, M. "Swarm Computing: The Emergence of a Collective Artificial Intelligence at the Edge of the Internet." IntechOpen.
- [8] HYPER-AI Project. "Hyper-Distributed Artificial Intelligence Platform for Network Resources Automation and Management." EU Cloud Edge IoT Wiki.
- [9] Gartner. "What is a Digital Immune System and Why Does It Matter?" Gartner Insights, 2022.
- [10] PingCAP. "Introduction to Leader Election in Distributed Systems."
- [11] Design Gurus. "How does leader election work in distributed clusters and why is it needed?"
- [12] "Beyond Autonomy: A Dynamic Tiered AgentRunner Framework for Governable and Resilient Enterprise AI Execution." arXiv preprint, 2026.
- [13] Avahi AI. "Autonomy Threshold." AI Governance Glossary.
- [14] Baumann, P. "Human in the Loop - The Responsible Choice?", citing Gartner's Human-AI Delegation Framework, 2026.
- [15] "Sovereign Edge Computing: What NATO-Aligned Organizations Need to Know." SNUC, 2026.
- [16] "Tactical Edge Computing." SNUC.
- [17] Peter, F. "Zero-Trust Architecture as a Building Block for Digital Sovereignty."
- [18] Boonma, J., and Rujipattanapong, S. "On the Study of the Effectiveness of iSCSI Protocols on the Representative Applications." Proceedings of ITC-CSCC 2007, Busan, Korea, July 8–11, 2007, pp. 979–980.

Author: Mr. Jakkrit Boonma | Email: jaggid.boonma@yahoo.co.th | Tel: +66 081-007-4833

© 2026 Jakkrit Boonma. All Rights Reserved. Ownership and copyright in this work belong to Mr. Jakkrit Boonma from the date of public disclosure, under applicable copyright law.